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The Relationship Between Pluripotency and Mitochondrial DNA Proliferation During Early Embryo Development and Embryonic Stem Cell Differentiation

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Abstract Pluripotent blastomeres of mammalian pre-implantation embryos and embryonic stem cells (ESCs) are characterized by limited oxidative capacity and great reliance on anaerobic respiration. Early pre-implantation embryos and undifferentiated ESCs possess small and immature mitochondria located around the nucleus, have low oxygen consumption and express high levels of glycolytic enzymes. However, as embryonic cells and ESCs lose pluripotency and commit to a specific cell fate, the expression of mtDNA transcription and replication factors is upregulated and the number of mitochondria and mtDNA copies/cell increases. Moreover, upon cellular differentiation, mitochondria acquire an elongated morphology with swollen cristae and dense matrices, migrate into wider cytoplasmic areas and increase the levels of oxygen consumption and ATP production as a result of the activation of the more efficient, aerobic metabolism. Since pluripotency seems to be associated with anaerobic metabolism and a poorly developed mitochondrial network and differentiation leads to activation of mitochondrial biogenesis according to the metabolic requirements of the specific cell type, it is hypothesized that reprogramming of somatic cells towards a pluripotent state, by somatic cell nuclear transfer (SCNT), transcription-induced pluripotency or creation of pluripotent cell hybrids, requires acquisition of mitochondrial properties characteristic of pluripotent blastomeres and ESCs.

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Introduction

Mitochondria are present in all eukaryotic cells except red blood cells and some protozoans. They play an important role in many cellular functions including pyrimidine biosynthesis, ion homeostasis, fatty acid oxidation (or β-oxidation) and cell signalling. However, one of the mitochondrion's key functions is the generation of adenosine triphosphate (ATP), the cell's major source of energy [1]. This results from the products of glycolysis, the citric acid cycle (or Krebs cycle) and fatty acid oxidation being further metabolised through the process of oxidative phosphorylation (OXPHOS) to produce 32 molecules of ATP. This aerobic process is far more efficient than anaerobic respiration, which only produces two molecules of ATP per molecule of glucose [2, 3].

Undifferentiated ESCs have low numbers of mitochondria and copies of mitochondrial DNA (mtDNA) [4–6] and restricted oxidative capacity [4, 5, 7] whereas fully differentiated cells vary considerably in their number of mitochondria and mtDNA copies per cell [8, 9], oxygen consumption and ATP production through OXPHOS [10–12]. Therefore, it is necessary for the elementary mitochondrial network of undifferentiated ESCs to be differentiated into the more active and mature mitochondrial networks that allow synthesis of greater levels of ATP and maintenance of differentiated cell homeostasis [2]. Indeed, the expression of nuclear-encoded and mitochondrial-encoded electron transport chain (ETC) subunits [5, 6], number of mitochondria and mtDNA copies/cell [4–6], the levels of oxygen

consumption and ATP production [4] through OXPHOS increase during the first 6 days of ESC differentiation. Moreover, these events which are related to the activation of aerobic metabolism seem to follow loss of pluripotency and cell fate commitment [6]. Here, we review the events of mtDNA transcription and replication that take place during oogenesis, pre-implantation development and ESC differentiation. In addition, we explore the relationship between pluripotency and the status of mitochondrial maturation and differentiation.

Role of Mitochondria

Aerobic production of ATP in the mitochondria is a complex process which requires concerted metabolism through glycolysis, fatty acid oxidation, the citric acid cycle and OXPHOS in the ETC [13]. Glycolysis takes place in the cytoplasm to metabolise glucose into pyruvate, nicotinamide adenine dinucleotide (NADH) and ATP [14]. In the mitochondria, acetyl-CoA is produced from the pyruvate derived from glycolysis and from fatty acid oxidation. Acetyl-CoA enters the citric acid cycle to further produce NADH, flavin adenine dinucleotide ($FADH_2$) and ATP [13, 15]. The NADH and $FADH_2$ produced during glycolysis, fatty acid oxidation and in the citric cycle are used by the ETC complexes located in the mitochondrial inner membrane to create an electrochemical potential between the mitochondrial matrix and the intermembrane space. Complex V (ATP synthase; ATPase) then uses the energy stored in the electrochemical gradient to produce ATP (Fig. 1) [1].

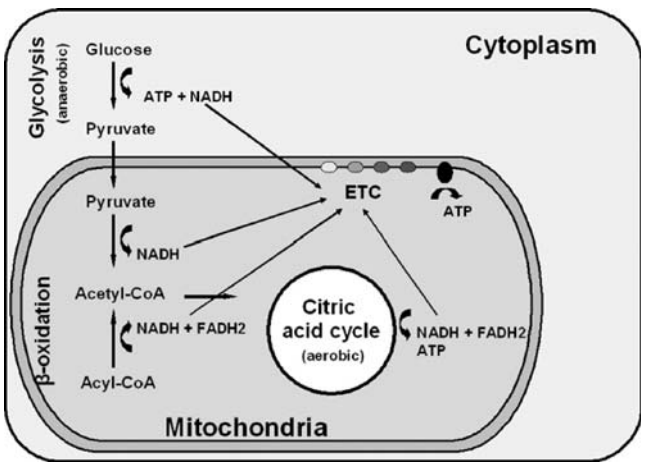


Fig. 1 Schematic representation of cellular respiration. During glycolysis, glucose is metabolised to produce pyruvate, NADH and ATP. In aerobic conditions, pyruvate produced from glycolysis is oxidized in the citric acid cycle resulting in production of further NADH and $FADH_2$. The electrons of NADH and $FADH_2$ are donated to the ETC to drive the synthesis of another 32 molecules of ATP through OXPHOS

The ETC is composed of five multiprotein complexes, Complex I (NADH dehydrogenase; ND), Complex II (succinate dehydrogenase; SD), Complex III (cytochrome c reductase), Complex IV (cytochrome c oxidase; COX) and Complex V and two electron carriers, coenzyme Q (CoQ) and cytochrome c (CytC; Fig. 2). The high energy electrons derived from NADH and $FADH_2$ are transferred into Complexes I and II, respectively. The electrons are then transferred from Complexes I and II to the lipid-soluble carrier coenzyme Q which in turn donates electrons to Complex III. Cytochrome c is the next electron acceptor and donates electrons to Complex IV. In the final step, the electrons are accepted by O_2 which results in the production of H_2O. Complexes I, III and IV use the energy released from the transfer of electrons to pump protons out of the mitochondrial matrix into the mitochondrial intermembrane space thus generating potential energy in the form of a pH gradient and an electrical potential across the mitochondrial membrane. The protons return to the mitochondrial matrix through Complex V which uses the energy stored in the electrochemical gradient to produce ATP from adenosine diphosphate (ADP) and inorganic phosphate (Fig. 2) [1, 16].

Mitochondrial DNA

Although the ETC complexes are mainly encoded by nuclear DNA (nDNA), Complexes I, III, IV and V have at least one subunit encoded by mtDNA [17]. Apart from 22 transfer RNAs (tRNAs) and two ribosomal RNAs (rRNAs), which are required for mitochondrial protein synthesis [18], mtDNA encodes seven subunits of Complex I (ND 1 to 6 and ND4L), one subunit of Complex III (cytochrome b; CytB), three subunits of Complex IV (COX I to III) and two subunits of Complex V (ATPase6 and ATPase8; Fig. 3) [17]. As the ETC complexes are encoded by both nDNA and mtDNA, a functional ETC requires well coordinated interaction between the nucleus and the mitochondria to ensure appropriate levels of transcription and translation of all ETC subunits and allow complete and functional assembly of the ETC [19, 20].

MtDNA is a double stranded circular molecule composed of a heavy (H) and a light (L) strand. The H-strand contains a higher G+C content and can therefore be separated from the L-strand on denaturing caesium chloride gradients based on density [21]. The H-strand encodes the 12S and 16S rRNAs, 14 tRNAs and 12 subunits of the ETC complexes. The L-strand encodes the remaining eight tRNAs and ND6 (Fig. 3) [20]. The size of mtDNA varies amongst individual species. For example, human mtDNA is 16,569 base pairs (bp) [17] in size, bovine is 16,338 bp [22], ovine is 16,616 bp [23] whereas murine mtDNA is 16,295 bp [24].

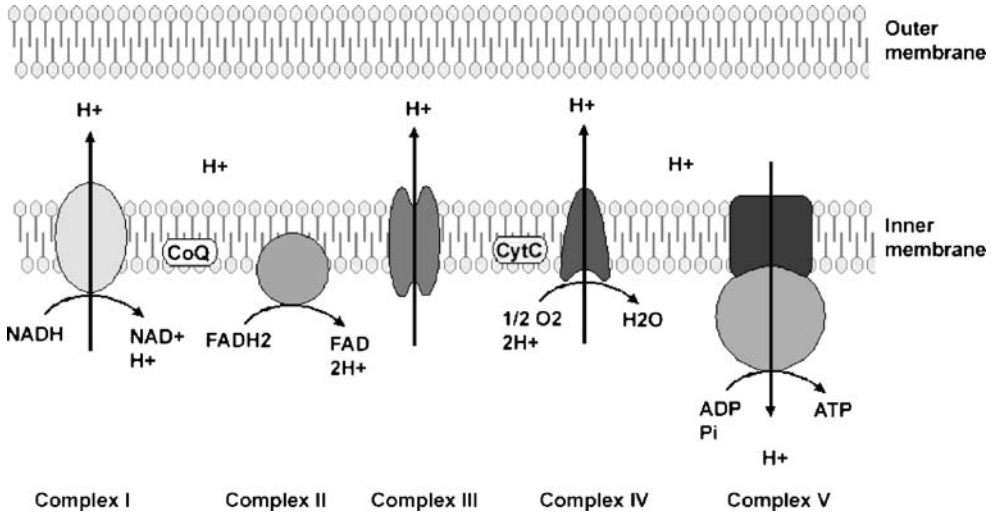


Fig. 2 The ETC produces the vast majority of ATP within a cell through OXPHOS. The ETC is composed of nDNA and mtDNA-encoded subunits organised into five enzyme complexes. NADH and $FADH_2$ donate electrons to Complexes I and II, respectively. The electrons flow through the first four complexes of the ETC and the energy released during the process is used by Complexes I, III and IV to pump protons across the mitochondrial inner membrane. The mitochondrial membrane potential that is generated is used by Complex V to produce ATP. *Complex I* (ND); Coenzyme Q (CoQ); *Complex II* (SD); *Complex III* (cytochrome c reductase); Cytochrome C (CytC); *Complex IV* (COX); *Complex V* (ATPase)

MtDNA is an extremely compact molecule as it contains only two non-coding regions and no introns between adjacent genes. The main non-coding region is the displacement loop region (D-loop) which possesses the origin of replication for the H-strand (O_H), the promoters for H- and L-strand transcription (HSP1, HSP2 and LSP) and a series of conserved sequence blocks (CSB) [17]. The second non-coding region is a 30 bp sequence which houses the origin of replication for the L-strand (O_L) and is located at about two thirds of the way round the genome from O_H [17]. Despite the absence of introns in mtDNA, the genes encoding the ETC subunits and rRNA are in most cases flanked by genes encoding tRNAs [25]. In the H-strand, sequences coding for tRNAs are not present between the genes ATPase6 and ATPase8, ND4L and ND4 and CytB and ND6 [25]. Another characteristic related to the compact organization of mtDNA is that the ATPase6 and ATPase8 genes overlap by 46 nucleotides and ND4 and ND4L genes overlap by seven nucleotides [25]. Moreover, in several genes, the termination codon is not encoded in the mtDNA, instead it is generated during post-transcriptional polyadenylation of the corresponding mRNA [25].

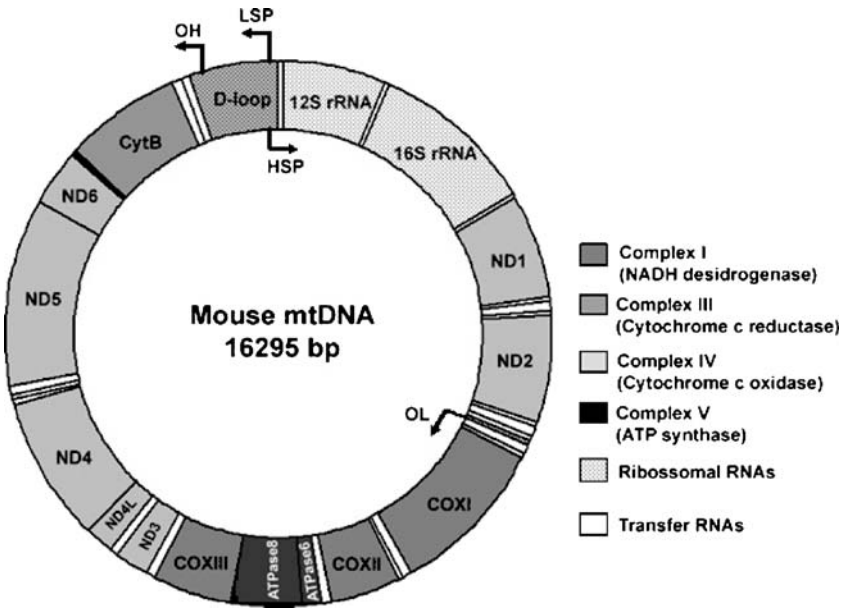


Fig. 3 The mtDNA genome according to the murine sequence [24]. mtDNA encodes seven subunits of Complex I (ND 1 to 6 and ND4L), one subunit of Complex III (CytB), three subunits of Complex IV (COX I to III) and two subunits of Complex V (ATPase6 and ATPase8), two rRNAs (12S and 16S rRNAs) and 22 tRNAs. The control region (*D-loop*) includes the H-strand promoter region (*HSP*), the L-strand promoter region (*LSP*), and the origin of H-strand replication (O_H). The second regulatory region, located between ND2 and COXI genes, contains the origin of L-strand replication (O_L)

MtDNA Transcription and Replication

Replication of mtDNA is a transcription dependent event as RNA transcripts initiated at the LSP are required for the initiation of mtDNA replication at the O_H [26]. Added to this, mtDNA transcription is reliant on nDNA-encoded transcription factors which are translocated to the mitochondria. The mitochondrial specific transcription factors include mitochondrial transcription factor A (TFAM) [27, 28], mitochondrial RNA polymerase (mtRNApol) [29], mitochondrial transcription factor B1 (TFB1M) and B2 (TFB2M) [30, 31] and mitochondrial transcription termination factor (mTERF) [32, 33].

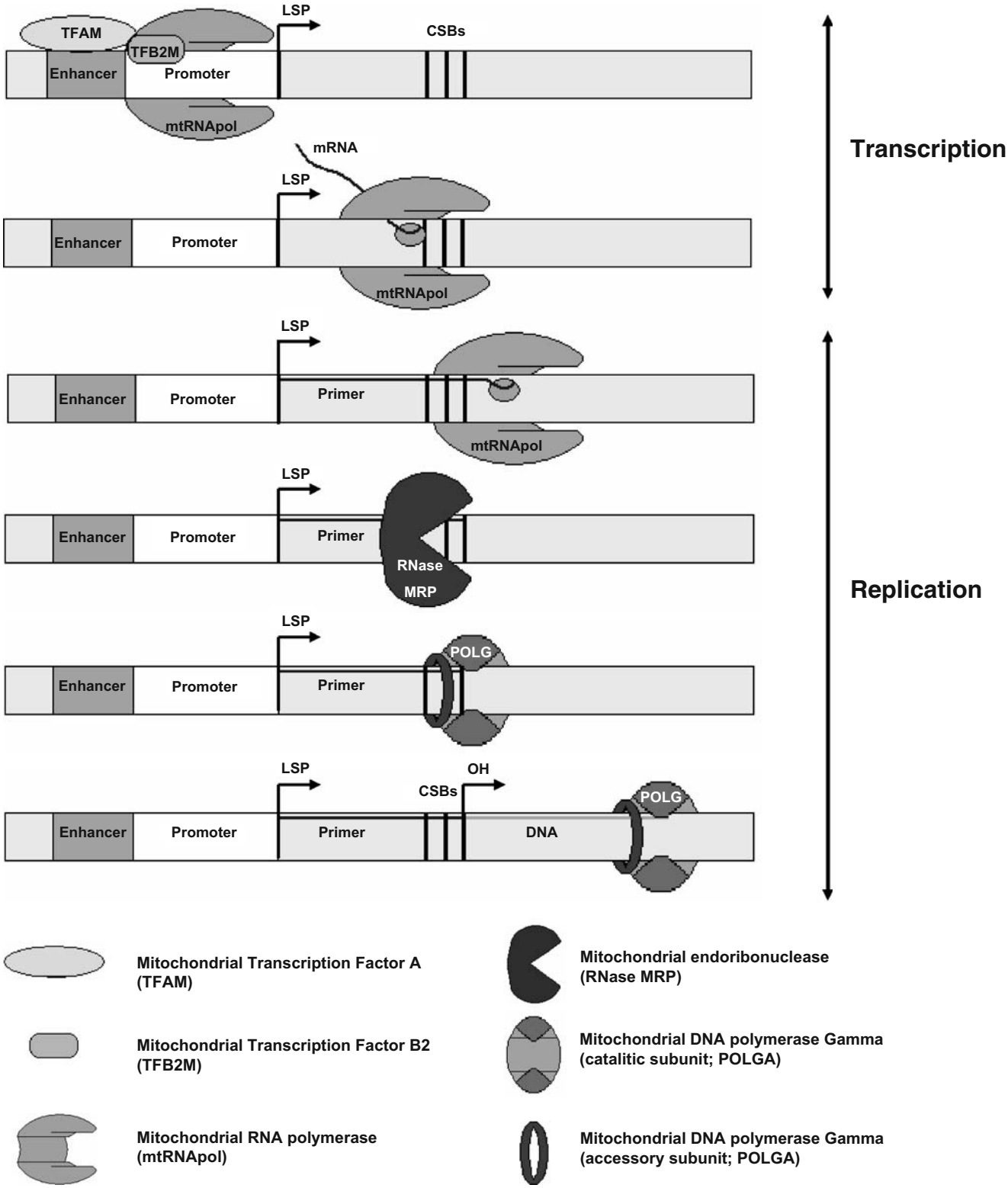
TFAM is a 25 KDa protein composed of two tandem high mobility group (HMG) box domains separated by a 27-amino-acid residue linker region and followed by a 25 amino acid residue carboxyl-terminal tail [34, 35]. TFAM's carboxyl-terminal tail recognises specific sequences of the enhancer sites directly upstream of LSP and HSP to induce structural alterations and possibly unwind the promoter region [34, 36]. At this stage, the accessory mitochondrial transcription factors TFB1M or TFB2M are capable of interacting with both TFAM and mtRNApol (Fig. 4) [30, 37]. mtRNApol, a 129 KDa protein, then recognises a sequence located between nucleotides +10 and −4 of the LSP through its promoter specificity region to initiate transcription of mtDNA [38, 39]. MtDNA transcription from HSP1 generally stops at a termination site just downstream of the two rRNA genes mediated by mTERF to form a truncated transcript that encodes the 12S and 16S rRNAs [32, 33]. Initiation of mtDNA transcription from HSP2 and LSP typically produces nearly full-genome length RNA [33, 40]. The long polycistronic primary transcripts coding for the H-strand genes give rise to the mature rRNA, mRNAs and tRNAs after cleavage at the endonucleolytic sites. Accurate processing of the H-strand polycistronic product is possible as the tRNA sequences acquire a cloverleaf structure that act as signals at the sites of cleavage allowing precise excision by the endonucleases [25, 41]. In the cases where no tRNAs are flanking the genes, it is likely that some RNA sequences acquire a tRNA-like secondary structure which can also be detected by the processing enzymes [25]. Similarly, endonucleolytic cleavage of the L-strand polycistronic transcripts generate the mature mRNA coding for ND6 and all the tRNAs encoded by the L-strand [20].

MtDNA is replicated by the mitochondrial-specific DNA Polymerase Gamma (POLG) [19, 21]. Mammalian POLG is a heterotrimer enzyme composed of one catalytic subunit (POLGA; 140 KDa) and two accessory subunits (POLGB; 55 KDa) [42]. POLGA contains the DNA polymerase, 3′→5′ exonuclease and 5′-deoxyribose phosphate (dRP) lyase domains [43, 44] whereas POLGB is required for recognition of the replication primer, increased polymerase and exonuclease activities and stimulates tight binding of the POLG complex to mtDNA [42, 45, 46]. Initiation of mtDNA replication begins with the transcription of an RNA primer from LSP [47], which necessitates the recruitment of TFAM, TFB1M or TFB2M and mtRNApol to transcribe the replication primer. Unlike with transcription, during initiation of mtDNA replication, the RNA sequence transcribed from the LSP remains hybridized to the L-strand in the CSBs [36]. The CSBs are conserved amongst all vertebrates and important for stabilizing the nascent triple-stranded DNA–RNA hybrid upstream of O_H (Fig. 4) [36, 48]. Then, the mitochondrial RNA processing endoribonuclease (RNase MRP), which is only active during mtDNA replication, cleaves the RNA sequence within the hybridized region to generate the mature RNA primer necessary for replication [36]. At this point, the mitochondrial single-stranded DNA-binding protein (mtSSB) stabilises the triplex structure [49], stimulates the activity of the mtDNA helicase, Twinkle [50], and stabilises the single-stranded mtDNA regions [51]. Twinkle is required to unwind the double-stranded mtDNA and for the progression of the mtDNA replication fork [50]. Supported by Twinkle and mtSSB, POLG proceeds to replicate the H-strand through extension of the RNA primer [52, 53].

Fig. 4 Schematic representation of the mammalian D-loop and the mtDNA transcription and replication factors during initiation of mtDNA transcription from LSP and mtDNA replication from O_H. TFAM binds specifically to the enhancer of LSP through its carboxyl-terminal tail and induces structural changes which expose the promoter region. TFB2M (or TFB1M) is able to interact with both TFAM and mtRNApol recruiting mtRNApol to the promoter region. MtRNApol binds to specific nucleotide sequences within the LSP to initiate mtDNA transcription. MtDNA transcription can then progress around the mtDNA genome to produce a long polycistronic transcript or produce an RNA primer which is processed by the RNase MRP at the CSBs. POLG uses the RNA primer to initiate replication of the H-strand. Although POLGA comprises the DNA polymerase domain, POLGB is also required for mtDNA replication as it increases considerably the processivity of the enzyme by stimulating polymerase activity and tight binding to the mtDNA genome

Additional Roles of TFAM, TFB2M and POLG

Apart from being involved in mtDNA transcription and replication, TFAM, TFB2M and POLG are also required for mtDNA packaging [54], translation [55] and repair [56], respectively. Although, TFAM's carboxyl-terminal tail is associated with specific binding to the promoter region and activation of transcription, TFAM is also able to bind, bend and unwind mtDNA without sequence specificity due to its two HMG box domains [34, 54]. This ability to bind


LSP
TFAM
TFB2M
CSBs
Enhancer
Promoter
mtRNApol
mRNA
Transcription
Primer
RNase
MRP
Replication
POLG
OH
DNA
Mitochondrial Transcription Factor A (TFAM)
Mitochondrial Transcription Factor B2 (TFB2M)
Mitochondrial RNA polymerase (mtRNApol)
Mitochondrial endoribonuclease (RNase MRP)
Mitochondrial DNA polymerase Gamma (catalitic subunit; POLGA)
Mitochondrial DNA polymerase Gamma (accessory subunit; POLGA)

mtDNA regardless of sequence specificity and its abundant presence [49, 54, 57] with very few molecules present in an unbound form in the mitochondrial matrix [54, 58] suggest that TFAM also packages mtDNA. In support of this, mtDNA depletion, following treatment with ethidium bromide, is accompanied by a decrease in TFAM levels whilst repopulation of mtDNA in the absence of ethidium bromide is followed by an increase in TFAM transcript levels [59, 60]. Furthermore, TFAM appears to bind mtDNA as a homodimer and is capable of fully compacting several DNA molecules together to form nucleoid structures [57].

TFB1M and TFB2M share high sequence homology with bacterial rRNA methyltransferases, enzymes involved in methylation of two adenine residues of the bacterial 16S rRNA during ribosome biosynthesis [30]. However, this rRNA methyltransferase activity is not related to the initiation of mtDNA transcription as point mutations that eliminate the activity of the methyltransferase motifs do not affect the ability of TFB1M to stimulate transcription *in vitro* [37]. In fact, TFB1M and TFB2M are dual-function proteins that not only support mtDNA transcription but also act as rRNA methyltransferases during translation of mitochondrial proteins. Indeed, TFB1M has been shown to methylate the conserved stem-loop in the bacterial 16S rRNA and the homologous sequence in the human mitochondrial 12S [55].

The exonuclease domain of POLGA is required for exonucleolytic proofreading which, when associated with high nucleotide selectivity, confers high fidelity DNA synthesis [61]. Indeed, expression of an exonuclease-deficient POLG in cultured human cells resulted in the accumulation of point mutations in mtDNA [62]. The 5′-dRP lyase domain is involved in base excision repair of the mtDNA genome [44]. The damaged base is firstly removed by DNA glycosylase and apurinic/apyrimidinic endonucleases and POLGA then excises the dRP residue through the 5′-dRP lyase domain and incorporates the missing nucleotide into the gap generated. DNA ligase catalyses the formation of a covalent phosphodiester bond between the 3′ hydroxyl end of the newly incorporated nucleotide and the 5′ phosphate end of the adjacent nucleotide to generate continuous mtDNA strands [56]. POLGB has also been shown to increase the efficiency of mtDNA repair by stimulating 5′-dRP lyase activity and increasing the ability of POLG to locate the damaged DNA [63].

Mitochondrial Diseases

Mitochondrial related diseases encompass a wide range of symptoms, severities and outcomes [64]. Mitochondrial disease can be inherited in Mendelian fashion as a result of mutations to nDNA genes [65] or be maternally inherited resulting from transmission of mutated mtDNA [66]. Mitochondrial diseases caused by mutation to nuclear genes are often associated with impaired mtDNA transcription and replication. For example, mutations in the PolgA gene are associated with several mitochondrial diseases, including dominant and recessive forms of progressive external ophthalmoplegia [67, 68], Alpers syndrome [69], Parkinsonism [70], ataxia–neuropathy [70], premature menopause and male infertility [71]. Such diseases, commonly originating through a mutation in the polymerase and/or exonuclease domains of POLGA, result in improper mtDNA replication and/or failure to proofread the newly synthesised mtDNA [72]. Subsequent mtDNA depletion and/or accumulation of mtDNA mutations inevitably lead to reduced oxidative capacity, low ATP production and the appearance of clinical symptoms [66]. Similarly, loss of function to POLGB also results in similar syndromes [73, 74].

MtDNA depletion syndromes are also frequently associated with decreased expression of TFAM. For example, low levels of TFAM protein were detected in skeletal muscle, liver, kidney, brain and heart of patients of infantile mitochondrial myopathy [75], skeletal muscle of patients with fatal mitochondrial myopathy [59], mitochondrial encephalomyopathy [76]. Although, mtDNA depletion was correlated with downregulation of TFAM levels in all these syndromes, the original cause of the diseases is not clear. This is because prolonged exposure to decreased expression of TFAM is likely to result in mtDNA depletion but mtDNA depletion can also dictate the levels of TFAM because the protein is not stable and is prone to degradation when unbound to mtDNA [75]. More recently, chimeric mice have been created through crosses of wild type mice with conditional knockout mice in which the Tfam gene is inactivated after birth in neurons of the neocortex and medial parts of the hippocampus [77, 78]. It was shown that inactivation of the Tfam gene results in the appearance of respiratory chain-deficient neurons and that the occurrence of less than 20% of these respiratory chain-deficient neurons in the forebrain are sufficient to cause clinical symptoms [78]. Furthermore, premature death of the chimeric mice occurred when the percentage of respiratory chain-deficient neurons in the forebrain was higher than 60% [78].

Mitochondrial diseases caused by point mutations to protein-coding genes of mtDNA include Leber's hereditary optic neuropathy (LHON) [79]; neurogenic weakness, ataxia, retinitis pigmentosa (NARP) [80]; mitochondrial encephalomyopathy [81]; and a combination of mitochondrial associated symptoms associated with mutation to nucleotide 8993 [82]. Mutation to the tRNA and rRNA genes cause myoclonic epilepsy and ragged red fibre disease (MERRF) [83]; progressive epilepsy with muscle weakness [84]; mitochondrial myopathy, encephalopathy, lactic acidosis and stroke-like episodes (MELAS) [85, 86];

and deafness [87]. In addition, mitochondrial diseases can be caused by large deletions of the mtDNA molecule, as for example Kearns–Sayre syndrome (KSS). Although a variety of mtDNA deletions differing in size and location are often found in KSS patients, one-third of all patients examined harbour a 4,977 bp deletion extending downstream of the 5′end of the ATPase8 gene to upstream of the 3′end of the ND5 gene [88]. This 'common' deletion in KSS patients is likely to arise from homologous recombination, as previously reported in lower eukaryotes and plants [88].

The phenotypic onset of maternally inherited mtDNA mutations will depend on the relative proportion of wild type and mutant mtDNA with over 85% of mutated mtDNA being normally required for the clinical disease phenotype to be expressed [89]. Nevertheless, the occurrence of phenotypic expression of mitochondrial diseases often results in dramatic consequences with 46% of the patients dying in the early ages of childhood [90]. As mtDNA is highly vulnerable to mutations due to the high levels of reactive oxygen species (ROS) and reduced antioxidant defences in mitochondria [91, 92], high levels of deleterious mtDNA mutations would be expected. Although analysis of the distribution of mutated mtDNA in the offspring of carrier mothers have shown that there is no strong selection against oocytes containing mutated mtDNA and that the risks of inheritance of the pathogenic mtDNA depend on its levels in the oocyte [93], recent data suggest that the transmission of deleterious mtDNA mutations depends on the severity of the mutation [94, 95]. Indeed, analysis of oocytes from chimeric mice carrying mild mtDNA mutations (mutation in the COXI gene that reduces the activity of Complex IV by 50%) and severe mtDNA mutations (mutation in the ND6 gene that completely inactivates Complex I) showed that the severe mtDNA mutation was negatively selected for during oocyte maturation resulting in complete elimination in successive generations whilst the mild mtDNA mutation was continuously transmitted [94]. Similar results were observed when females expressing a proofreading-deficient PolgA were mated with wild-type males [95]. Here, the mtDNA mutations in the protein-coding genes causing amino acid changes were much less represented in successive generations than "silent" mutations suggesting the existence of a process that prevents transmission of deleterious mtDNA mutations [95].

Pluripotency During Early Embryo Development and in ESCs

Early embryonic development in mammals is characterised by a progressive reduction in the developmental potential of embryonic cells [96, 97]. The fertilized egg (zygote) and the early blastomeres are totipotent, as they can give rise to both embryonic and extra-embryonic tissues [98, 99]. Pluripotency is indicative of the inner cell mass (ICM) and the epiblast of the mammalian embryo and undifferentiated ESCs and refers to the potential of these cells to form all three embryonic germ layers (endoderm, mesoderm and ectoderm) and consequently all cell types in the body [100, 101].

Pluripotency in early cleavage stage embryos is regulated mainly by maternally inherited pluripotency transcription factors, including OCT4 [102] and SOX2 [103], as well as epigenetic factors that regulate histone modifications required for transcriptional activation of the embryonic genome [99]. In mouse, embryonic expression of Oct4 is initiated at the late four-cell stage but becomes restricted to the pluripotent cells of the ICM at the blastocyst stage [102, 104]. Similarly, expression of the transcription factor Sox2 was also detected in all cleavage stages and in the ICM and epiblast lineages [103]. Consistent with their major role in maintenance of pluripotency during pre-implantation development, failure to express Oct4 or Sox2 results in loss of pluripotency in the cells of the ICM and differentiation into the trophectoderm, which leads to embryonic death at E4.5 [105] and E7.5 [103], respectively.

Nanog transcripts are not detected in the mature oocyte, zygote or early cleavage stages but are first detected in the inner cells of the compacted morula [106]. At the blastocyst stage, Nanog expression is also restricted to the ICM [106, 107]. Although homozygous Nanog mutant embryos give rise to an ICM, they fail to maintain pluripotency in the cells of the epiblast which instead differentiate into primitive endoderm cells resulting in embryonic death prior to E5.5 [107]. Therefore, the homeodomain protein NANOG is likely to cooperate with OCT4 and SOX2 during late pre-implantation and early post-implantation embryogenesis to maintain the pluripotency of the epiblast cells [106].

As ESCs are derived from the mammalian blastocyst [100, 101], it is not surprising that the major regulators of pluripotency in pre-implantation embryo development are also required for maintenance of pluripotency and self-renewal in ESCs. Indeed, Oct4, Sox2 and Nanog are highly expressed in undifferentiated mouse and human ESCs [106, 108, 109]. Moreover, siRNA knockdown of Oct4 [110, 111] and Sox2 [112] or conditional inactivation of an Oct4 transgene [108] in mouse and human ESCs led to results similar to those observed in homozygous Oct4 [105] and Sox2 [103] knockout mice, with reduction of OCT4 or SOX2 levels resulting in loss of the undifferentiated state and increased expression of trophectoderm markers. Similarly, siRNA knockdown of Nanog in mESCs [111] results in loss of pluripotency and differentiation of the extraembryonic endoderm as in homozygous Nanog knockout mice [107].

Also involved in the maintenance of pluripotency are the Oct4-related genes Dppa5 (also called ESG1), Pramel7 and

Ndp52l1 [113]. Expression of these genes is restricted to the pluripotent cells of pre-implantation embryos, undifferentiated ESCs and primordial germ cells (PGCs) [113–115]. Interestingly, overexpression of Nanog and Pramel7 maintains self-renewal and pluripotency in the absence of LIF/STAT3 stimulation [106, 115] whereas overexpression of Oct4 induces differentiation of mESCs into extraembryonic primitive endoderm and mesoderm suggesting that precise levels of Oct4 expression are required for maintenance of self-renewal and pluripotency [108].

Association Between Pluripotency and Mitochondrial Biogenesis During Early Embryo Development

During differentiation of PGCs into mature oocytes, the number of mitochondria and mtDNA molecules is greatly increased [116, 117]. Although there are variable estimates, it is suggested that human and murine PGCs contain between 10 and 100 mitochondria [93, 118, 119] whereas mature oocytes will contain between 75,000 to 100,000 based on the principle that there are 1 to 2 mtDNA copies per organelle [93]. Indeed, mature murine oocytes have been reported to have mean values of 114,000 [120], 157,000 [119] and 249,000 mtDNA molecules [121] whilst mature human oocytes contain mean values of 138,000 [117], 193,000 [122], 250,454 [123] and 314,000 mtDNA molecules [124]. As the number of mtDNA copies in the mature oocyte varies considerably, it has been suggested that this is related to oocyte fertilization and embryo development rates. To this extent, a critical threshold of at least 100,000 mtDNA copies has been proposed as an indicator of oocyte competence [123, 125, 126]. Decreased mtDNA content and poor oocyte fertilisation rates may result from deficient oocyte maturation as immature mouse oocytes have considerably lower mtDNA copy number than mature oocytes [120, 126].

The dependence of successful embryo development on oocytes containing a large number of mtDNA molecules arises as mitochondria in the embryo are exclusively inherited from the oocyte, with no sperm mitochondria being detected in mouse embryos after the two-cell stage [127] and mtDNA replication post-fertilization most likely not taking place until the blastocyst stage [120, 128, 129]. Although a small mtDNA replication event has been reported in pronuclear (one-cell) and two-cell stage mouse embryos, a concomitant event of mtDNA degradation must have occurred as no increase in number of mtDNA copies was detected [130]. Therefore, during pre-implantation stages of embryo development, each newly divided blastomere contains fewer mtDNA copies after each embryonic cell division. Consequently, it is not surprising that the reliance on OXPHOS is decreased during mouse and human pre-implantation development and glycolysis provides the major source of ATP [131]. This led to the suggestion that the increase in glycolytic flux is likely to be important for the survival of blastomeres containing a limited number of mtDNA copies and restricted oxidative capacity but that elevated mtDNA content at fertilisation would still be vital to ensure that there are sufficient copies of mtDNA molecules per blastomere to allow transmission of mtDNA until mtDNA replication is reinitiated [125]. Indeed, it has also been suggested that a large number of mitochondria and mtDNA in the oocyte represent a genetic mechanism to ensure their even distribution to the blastomeres because the occurrence of low levels of mitochondria and mtDNA would more likely result in maldistribution throughout the embryo [93]. Also related to the decreasing oxidative capacity during pre-implantation stages of embryogenesis, it has been shown that embryos that develop from oocytes with ATP content being ≥2 pmol have a higher potential for continued embryogenesis and implantation than embryos that develop from oocytes containing less than 2 pmol of ATP/oocyte [132, 133]. Moreover, supplementation of mouse, human [134] and pig [125] oocytes with active mitochondria resulted in higher levels of ATP production and improved rates of embryo development suggesting that mitochondrial metabolism might also be important for determining oocyte competence to support embryo development [125, 134].

Although the expression of mtDNA transcription and replication factors and mtDNA-encoded genes is detected during the cleavage stages in a number of mammalian species including mouse, sheep and pig, the levels of expression before the morula stage were low and most likely associated with the generic expression of a whole host of genes during embryonic genome activation (EGA) [135]. For example, in mouse, upregulation of the mtDNA-encoded COXI and COXII was initially detected at the late two-cell stage [120], the same stage at which EGA takes place [136], and more consistent levels of mtDNA-encoded ETC subunits were only detected at the morula stage [120, 129]. In sheep, PolgA, PolgB and Tfam were also initially upregulated during the eight-cell to 16-cell stages [137], when the embryonic genome starts to be transcribed [138], but higher levels of these mtDNA transcription and replication factors were detected at the morula and blastocyst stages [137]. Similarly, in pig, PolgA, PolgB and Tfam were upregulated during EGA at the four- to eight-cell stage [139] but more consistent expression was detected at the morula and blastocyst stages [126]. Specifically though, PolGA expression was eliminated after EGA, unlike PolGB and TFAM. In cattle, the increase in the Tfam, COXI and CytB mRNA levels only took place at the morula stage of embryonic development [128]. Consequently, initiation of major mtDNA transcription and replication activities

seems to only take place at the late morula or blastocyst stages [126, 137].

Consistent with the activation of mtDNA transcription and replication at the later stages of pre-implantation development, mitochondria also start to differentiate into elongated organelles containing swollen cristae and acquire high membrane potential [140–142]. In addition, the levels of oxygen consumption and ATP production through OXPHOS are also clearly increased at the later stages of pre-implantation as a result of the activation of mitochondrial aerobic metabolism [141, 143, 144].

Nevertheless, mitochondrial differentiation and the increase in the levels of oxygen consumption and ATP production at the blastocyst stage are restricted to the cells differentiating into the trophectoderm lineage which, in turn, increases the number of mitochondria, levels of POLG and TFAM expression and amino acid turnover [126, 145]. The cells of the ICM continue to be pluripotent as shown by the expression of Oct4, Tra-1–60, Tra-1–81 and alkaline phosphatase [146] and remain relatively quiescent with persistently low levels of POLG and TFAM [126] and still rely on anaerobic metabolism [145]. In addition, the trophectoderm cells contain long and thin mitochondria whereas the mitochondria of ICM cells are still spherical [142]. Therefore, high levels of expression of pluripotency genes appear to be associated with low levels of expression of mtDNA transcription and replication factors whereas the first events of cellular differentiation during pre-implantation development seem to upregulate the expression of POLG and TFAM to activate mitochondrial aerobic metabolism which is restricted to the trophectoderm.

Mitochondria in Undifferentiated ESCs

Undifferentiated mouse and human ESCs, which are derived from the ICM, have a relatively undeveloped mitochondrial network and contain only a few mitochondria with poorly developed cristae [4, 5, 7, 147]. Moreover, undifferentiated ESCs seem to conserve the mitochondrial characteristics and patterns of cytoplasmic localization detected during early stages of mammalian embryo development [148]. Indeed, spherical and immature mitochondria located around the nucleus similar to those observed in mouse [149], cattle [133], pig [126] monkey [150] and human pre-implantation embryos [141] have also been found in human [4, 5] and mouse ESCs [6, 7], once more suggesting an association between pluripotency and the state of mitochondrial differentiation.

Undifferentiated mESCs tend to express low levels of PolgA, PolgB and Tfam and contain low mtDNA content [6]. Indeed, real time PCR analysis detected only 31 mtDNA copies/cell in R1 mESCs, 37 copies/cell in D3 mESCs [6] and 44 mtDNA copies/cell in CCE/R1 mESCs [151]. Similarly, undifferentiated hESCs contain low mRNA levels for Tfam and Tfb2m and undetectable levels of PolgA and Tfb1m transcripts [5]. Moreover, mouse and human ESCs show low levels of ATP production and express modest levels of antioxidant enzymes [4, 7] which relates to the increased expression of glycolytic enzymes and high glycolytic flux [7, 152] and decreased expression of ETC subunits [5] and citric acid cycle enzymes [7]. Also related to poor oxidative capacity, mESCs have been shown to have low oxygen consumption [152]. Furthermore, culture of hESCs in hypoxic conditions (3 to 5% O_2), similar to those detected before implantation and vascularization *in vivo* [153], has been shown to promote normal growth and support the maintenance of pluripotency of hESCs in comparison to the traditional culture conditions in 21% O_2 [154].

As previously mentioned, preservation of immature mitochondria with a perinuclear arrangement, reduced expression of OXPHOS enzymes and low metabolic activity in pre-implantation embryonic cells and ESCs has led to the suggestion that these mitochondrial properties might be important for the maintenance of pluripotency and self-renewal and should therefore be considered as another ESC marker [148, 155]. However, it seems that adult stem cells also share these mitochondrial properties as the perinuclear localization of elementary mitochondria has also been reported in human hematopoietic stem cells (HSCs) [156] and monkey [155] and human mesenchymal stem cells (MSCs) [157].

Association Between Loss of Pluripotency and Mitochondrial Biogenesis During Differentiation of ESCs

Although, anaerobic metabolism is sufficient to meet the energy demand of undifferentiated mouse and human ESCs [4, 7], successful differentiation of ESCs requires activation of mitochondrial aerobic metabolism [5, 7, 152]. The efficiency of ATP production through OXPHOS is far superior compared to glycolysis [2, 3] and thus the immature mitochondrial network of undifferentiated ESCs must be differentiated into a more active and mature network in order to promote the synthesis of greater levels of ATP and the maintenance of homeostasis in the differentiated cell [2]. Indeed, fully differentiated cells such as neurons and cardiomyocytes express high levels of nDNA- and mtDNA-encoded ETC subunits, produce ATP through OXPHOS [5, 7, 12] and contain enriched mtDNA content [8, 9].

Nevertheless, activation of aerobic production of ATP during ESC differentiation, as in mammalian embryo

development, requires the downregulation of pluripotency genes and commitment to a specific cell fate [6]. Indeed, downregulation of Dppa5, Pramel7, Ndp52l16 and Nanog (Fig. 5A) and the increase in the number of mESCs expressing mesoderm and neuroectoderm markers takes place during days 3 and 4 of differentiation in the presence of low levels of PolgA, PolgB and Tfam expression and minimal levels of mtDNA transcription and replication [6]. Interestingly, Oct4 is downregulated during days 1 to 4 but upregulated on days 5 and 6 of differentiation (Fig. 5A).

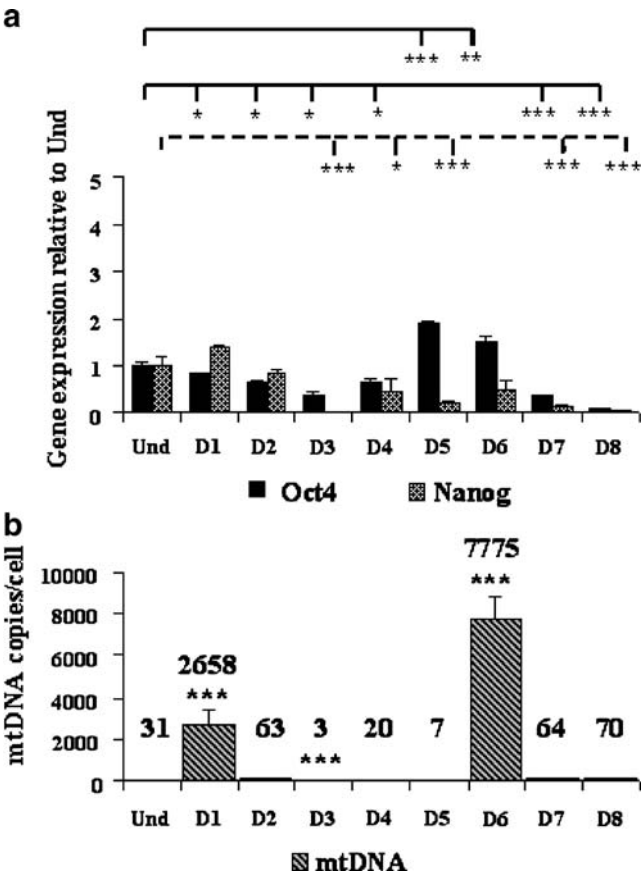


Fig. 5 Loss of pluripotency and mtDNA replication during differentiation of mESCs. **a** Nanog was upregulated on day 1 (1.40 fold; $P>0.05$) but downregulated to undetectable levels ($P<0.001$) on day 3 and to 44% ($P<0.05$) of the levels for the undifferentiated mESCs on day 4 of differentiation. On days 5, 6, 7 and 8, the mRNA levels for Nanog were 24% ($P<0.001$), 49% ($P>0.05$), 17% ($P<0.001$) and 6% ($P<0.001$) of the levels for the undifferentiated mESCs, respectively. Oct4 was downregulated during days 1 to 4 by between 83% and 39% ($P<0.05$) of the levels for the undifferentiated mESCs, upregulated on days 5 (1.9 fold; $P<0.001$) and 6 (1.52 fold; $P<0.05$) but very significantly downregulated on days 7 (37%; $P<0.001$) and 8 (11%; $P<0.001$). **b** Undifferentiated R1 mESCs had 31 mtDNA copies/cell. Following spontaneous differentiation, the mtDNA copy number increased significantly above the undifferentiated R1 mESCs on days 1 (2,658 copies/cell; $P<0.001$) and 6 (7,775 copies/cell; $P<0.001$). At day 3, mtDNA copy number decreased to a minimum of three copies/cell ($P<0.001$) (adapted with permission from Facucho-Oliveira et al. 2007 [6]). *Bars* represent mean ± s.e.m.; significant differences between days are indicated (*$P<0.05$, **$P<0.01$, ***$P<0.001$)

Defined levels of Oct4 expression are required for the maintenance of pluripotency with either overexpression [108] or knockdown of Oct4 [110, 111] resulting in differentiation of extraembryonic trophectoderm or extra-embryonic primitive endoderm and mesoderm, respectively. Increased and decreased levels of OCT4 levels are believed to cause changes in the stoichiometry between OCT4 and the pluripotency transcription factors SOX2 and NANOG which in turn results in alteration of expression of their target genes [158, 159]. Perhaps as a result of this, mtDNA is then extensively replicated (Fig. 5B) [6], the number of mitochondria increased [4, 6, 7] and the expression of nDNA- and mtDNA-encoded ETC subunits clearly upregulated on day 6 of differentiation [4, 6].

In addition, differentiation of mouse and human ESCs results in changes in mitochondrial structure, morphology and patterns of cytoplasmic localization. Mitochondria migrate from the perinuclear regions into more dispersed cytoplasmic areas, acquire a tubular structure, numerous elongated cristae, dense matrices and high membrane potential suggesting the initiation of metabolic activity through OXPHOS [4, 5, 7]. Consistent with this, the expression of glycolytic enzymes decreases and the expression of ETC and citric acid cycle enzymes increases during cardiac differentiation of mESCs [7]. Furthermore, the levels of ATP production, oxygen consumption and cellular respiration are also increased whilst anaerobic production of ATP is reduced [4, 7].

Related to the activation of aerobic metabolism, the expression of the antioxidant enzymes Cu/Zn superoxide dismutase (Cu/Zn-SOD), glutathione peroxidase 1 (GPx1) and peroxiredoxins 1 and 2 (Prx1 and Prx2) dramatically increases during the first 7 days of hESC differentiation [4, 160]. Nevertheless, the intracellular levels of ROS are higher in differentiating ESCs than in undifferentiated ESCs due to the increase in OXPHOS metabolism [160]. However, an increase in ROS levels might have a role in cell signaling and regulation of proliferation and differentiation. Exposure to low levels of ROS has been reported to enhance ESC differentiation towards the cardiomyogenic and vascular lineages whereas continuous exposure to high levels of ROS results in inhibition of differentiation [160].

Comparative analysis of embryoid bodies (EBs) and early post-implantation embryos showed that expression and overall distribution of stage and lineage specific markers during EB differentiation strictly mirrors the expression patterns during early post-implantation development [161]. Spontaneously differentiated EBs at days 1 and 2 are equivalent to mouse embryos at E4.5 to E6.5, EBs at days 3 to 5 are equivalent to E6.5 to E7.0 and that EBs after day 6 are equivalent to E7.5 (Fig. 6) [161]. Moreover, the embryos at E4.5 and E6.5 are at pre-gastrulation stages which correlate with the expression of the pluripotency genes Nanog (Fig. 5A), Dppa5, Pramel7 and Ndp52l1 during

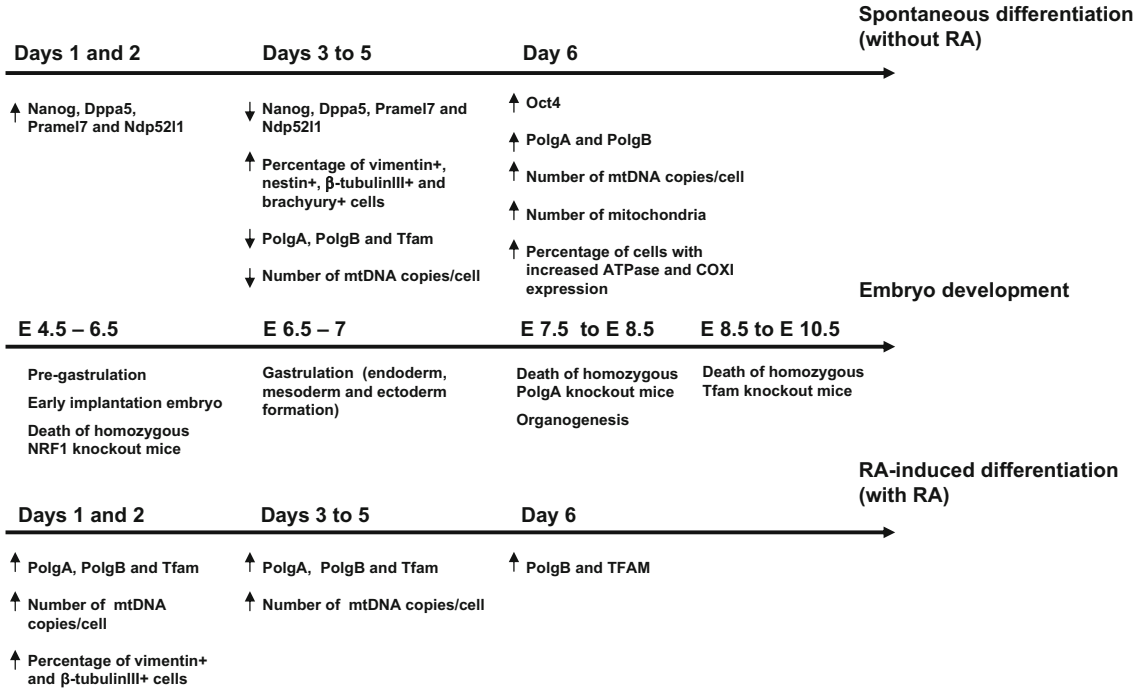


Fig. 6 Schematic representation of the events involved in mitochondrial biogenesis during spontaneous differentiation (without RA), RA-induced differentiation (with RA) and early stages of embryo development. Comparative analysis of pre-implantation and early post-implantation murine embryos and differentiation of mESCs shows a strong correlation between stages of embryo development and days of mESC spontaneous differentiation [161]. High levels of expression of pluripotency genes in mESCs coincided with the pre-gastrulation stages of embryo development. The loss of pluripotency and cell fate commitment during spontaneous differentiation of mESCs coincided with differentiation of the three embryonic germ layers during gastrulation. Activation of mitochondrial aerobic metabolism on day 6 of mESC differentiation [6] coincided with embryonic organogenesis and death of homozygous PolgA [162] and Tfam knockout mice [161, 163]. Unlike the spontaneously differentiated mESCs, the RA-induced mESCs increased the number of neuroectoderm committed cells on day 1. Moreover, the faster onset of differentiation in the RA-induced mESCs is likely to be related to the earlier upregulation of mtDNA transcription and replication factors and an increase in the number of mtDNA copies/cell in order to stimulate earlier activation of aerobic metabolism

days 1 and 2 of mESCs differentiation [6]. At E6.5 to E7.0 the embryos are in gastrulation as detected by the expression of endoderm, mesoderm and ectoderm markers [161] which in mESCs coincided with down-regulation of the pluripotency genes and upregulation of lineage specific markers during days 3 and 4 of differentiation (Fig. 6) [6]. At E7.5, mouse embryos are in the early stages of organogenesis [161], which coincides with the upregulation of Oct4 (Fig. 5A) and the increase in expression of the mtDNA replication factors and ETC subunits, number of mtDNA copies/cell and number of mitochondria on day 6 (Fig. 6) [4, 6]. Moreover, as homozygous PolgA and Tfam knockout mice die between E 7.5 and 8.5 and E 8.5 and 10.5, respectively (Fig. 6) [162, 163], it is very likely that the events of mtDNA replication described during ESC differentiation mirror the onset of mitochondrial biogenesis during early embryo development. Indeed, failure to replicate mtDNA on E7.5 would explain the severe mtDNA depletion and impaired OXPHOS which ultimately led to loss of organogenesis and embryonic death of homozygous PolgA and Tfam knockout mice.

Interestingly, mESCs differentiated in the presence of retinoic acid (RA; RA-induced differentiation) had higher levels of PolgA, PolgB and Tfam expression and higher numbers of mtDNA copies/cell than spontaneously differentiated mESCs (without RA) during days 1 to 5 of differentiation (Fig. 6) [6]. Exposure to RA is likely to have resulted in upregulation of PGC-1α expression [164], a co-activator of nuclear respiratory factors 1 and 2 (NRF1 and NRF2) [164, 165]. As NRF1 and NRF2 are transcription factors which activate and coordinate the expression of various mtDNA transcription and replication factors and nDNA- and mtDNA-encoded ETC subunits [166, 167], activation of NRF1 and NRF2 upon RA exposure could certainly upregulate the expression of PolgA, PolgB and Tfam and consequently induce mtDNA transcription and replication. Indeed, increased expression of various mtDNA-encoded ETC subunits has been reported following treatment with RA [168, 169]. However, direct stimulation of RA on mtDNA replication cannot be excluded as retinoic acid receptors (RARs) have been found in the mitochondria [169] and retinoic acid response elements

(RARE) identified in the regulatory region (D-loop) of the mtDNA genome [170, 171].

In addition, the RA-induced ESCs differentiate far more rapidly with significantly higher number of cells being committed to the neuroectoderm lineage on days 1 and 2 than their spontaneously differentiated counterparts (Fig. 6) [6]. Indeed, various pluripotency genes including Oct4 and Nanog have been shown to be more rapidly downregulated [172] and the expression of the neuronal transcription factors, Mash1 [173], Pax6 [174, 175] and Hoxa1 [172] and the signalling molecule Wnt1 [173, 176] more sharply increased during differentiation of mESCs in the presence of RA. The faster onset of differentiation and earlier increase in mtDNA copy number in the RA-induced ESCs further support the concept that loss of pluripotency, cell fate commitment and activation of aerobic mitochondrial metabolism are sequential events during ESC differentiation and *in vivo* embryo development.

Further evidence of the importance of mitochondrial biogenesis for embryo survival and successful differentiation of ESCs arises from homozygous NRF1 knockout mice [177]. Consistent with its critical role during mitochondrial biogenesis, homozygous disruption of NRF1 resulted in earlier developmental arrest than homozygous disruption of PolgA and Tfam, with defective mitochondrial membrane potential and mtDNA depletion causing embryonic death between E 3.5 and 6.5 (Fig. 6) [177]. In addition, exposure to mitochondrial poisons has also been shown to disrupt cardiac differentiation of mESCs. EBs treated with the ETC inhibition agents antimycin and rotenone fail to develop to complete cardiomyogenesis and showed deficient sarcomere formation, reduced numbers of beating areas, defective contractility, reduced volume and aberrant mitochondrial distribution [7, 178].

Transmission of mtDNA Following Somatic Cell Nuclear Transfer (SCNT)

Nuclear transfer (NT) is a technique which involves the injection of a donor nucleus or a whole cell into an enucleated recipient oocyte [179]. The donor cell can either be an embryonic cell (ECNT) or a somatic cell (SCNT). After artificial activation (electrically or biochemically), the reconstructed oocyte undergoes pre-implantation development in a manner similar to natural fertilisation. However, the efficiency of SCNT is very low, with reduced numbers of reconstructed oocytes developing until the blastocyst stage [180] and progressively fewer percentages of embryos implanting, surviving pregnancy and producing live offspring [181]. One of the causes contributing to the poor efficiencies of cloning and the high proportion of embryonic death is incorrect reprogramming of the donor cell nucleus. Indeed, comparison of the expression of the imprinted genes Igf2, Igf2r and H19 in various tissues of calves that died soon after birth and calves that survived until adulthood showed that expression in the live clones was mostly normal and comparable to the control animals whereas in the deceased animals the levels of expression were highly variable [182].

Another important consideration in SCNT is the pattern of mtDNA inheritance. Upon natural fertilization, IVF or intracytoplasmic sperm injection (ICSI), paternal mitochondria entering the oocyte are eliminated in intraspecific crossings and only the mitochondria and mtDNA derived from the oocyte are transmitted to the embryo [127, 183]. Elimination of paternal mitochondria by the oocyte is thought to be possible because sperm mitochondria are tagged with ubiquitin during early spermatogenesis and can therefore be distinguished from maternal mitochondria and selectively eliminated by proteolysis during the early cleavage stages of embryonic development [184]. However, in SCNT, the several hundred mitochondria from the donor cells that are introduced into the enucleated oocyte are not efficiently eliminated which can lead to offspring being either homoplasmic for recipient oocyte mtDNA (containing oocyte derived mitochondria only) or heteroplasmic resulting from transmission of both donor cell and recipient oocyte mtDNA [185, 186].

The persistence of two (or more) mtDNA populations within the embryonic cells is another possible explanation for the low efficiency of cloning. Differences in mtDNA sequence between donor cell and recipient oocyte are likely to give rise to proteins with slight differences in their respective amino acid sequences [185]. This is likely to result in inadequate interaction between the individual ETC subunits which might lead to reduced oxidative capacity and consequently embryonic or fetal death [187, 188]. In addition, failure or defective interaction of the donor nDNA-encoded mtDNA transcription and replication factors with the recipient oocyte's mtDNA would result in compromised mtDNA transcription and replication and most likely lead to embryonic arrest or to the appearance of mtDNA-depletion syndromes after birth. Supporting these concepts, brain, kidney and liver defects in sheep NT embryos have been associated with mitochondrial dysfunction and have been suggested as a common cause of death resulting from reproductive cloning [189]. Although, the levels of heteroplasmy have been reported to vary between 0% to 59% of the embryo's total mtDNA content [190], variable levels of donor mtDNA transmitted from the embryo are also found in different tissues of cloned animals [190, 191]. Varying levels of heteroplasmy have been related to quantity of donor cell cytoplasm that are injected during cloning and the number of mitochondria and mtDNA present in the donor cell [190]. However, more recent analysis of SCNT embryos generated

with non-depleted donor cells containing their full mtDNA complement ($mtDNA^{+}$) and donor cells depleted to residual levels of mtDNA ($mtDNA^{R}$) showed that donor cell mtDNA was still present in both SCNT embryos generated with $mtDNA^{+}$ and $mtDNA^{R}$ cells at the blastocyst stage [137, 192]. This arises as these embryos were unable to regulate expression of PolGA, PolGB and Tfam encoded by the donor cell nucleus and thus these mtDNA transcription and replication factors preferentially replicate their own mtDNA populations [137, 192]. Indeed, in the $mtDNA^{+}$ and $mtDNA^{R}$ cloned embryos, the transcript levels for the mtDNA transcription and replication factors was significantly higher than their IVF counterparts at the two-cell stage and the protein synthesized for POLG and TFAM persisted through to the 16-cell stage [137].

Pluripotency and Mitochondria in SCNT, Cell Fusion and Transcription-Induced Pluripotency

SCNT is based on the principle that the cytoplasm of the oocyte is capable of reprogramming the donor nucleus prior to the first cell division [179]. As each differentiated cell-type has its own specific gene expression profile, which relates to the cell's function and morphology [193], successful SCNT requires a complete reprogramming of the somatic cell nucleus to acquire the totipotency properties of the zygote nucleus. Many studies have focused attention on analysis of epigenetic modification on nDNA, including DNA methylation, histone modification and chromatin remodelling, during SCNT [194]. However, reprogramming of mitochondria or mtDNA has not been considered despite the clear similarities in mitochondrial morphology, patterns of cytoplasmic localization, expression of mtDNA transcription and replication factors and ETC subunits and overall metabolic activity between the pluripotent cells of pre-implantation embryos and undifferentiated ESCs.

Most somatic cells contain a well developed mitochondrial network composed of hundreds of mitochondria and even higher numbers of mtDNA copies/cell [8, 195–197] whilst the pluripotent cells of the embryo are characterised by small and immature mitochondria located in the perinuclear regions of the cytoplasm [126, 133, 141]. Therefore, donor cell mitochondria and mtDNA in heteroplasmic SCNT embryos are likely to require reprogramming upon activation of the reconstructed oocyte in order to acquire properties similar to the mitochondria in pre-implantation embryos. However, mature mitochondria with swollen cristae have been detected during the early cleavage stages of SCNT embryo development but not in the control embryos until the 16-cell stage suggesting that reprogramming of donor cell mitochondria in heteroplasmic SCNT embryos is somehow incomplete [198]. Failure to reprogram donor cell mitochondria is likely to result in embryos and ESCs derived from SCNT expressing mitochondria and numbers of mtDNA which are phenotypically characteristic of the donor cell type. Those cells might therefore lack the level pluripotency required to generate all cells of the body or contain mitochondria with insufficient plasticity to differentiate and fulfil the multitude of ATP requirements of all cell types.

Similar outcomes would also be expected in pluripotent cells derived from transcription-induced pluripotency or cell fusion of a somatic cell with ESCs. Induced pluripotency consists of reprogramming somatic cells by manipulating the expression of factors that regulate pluripotency [199, 200]. Human induced pluripotent stem cells (iPSCs) have been recently generated by overexpression of Oct4, Sox2, c-myc and Klf4 in adult human fibroblasts [201]. On the other hand, the creation of pluripotent cell hybrids through fusion of a pluripotent cell with a somatic cell is based on the potential of the pluripotent nucleus to reprogram the somatic nucleus [202]. Both iPSCs and pluripotent cell hybrids have gene expression profiles, patterns of methylation of the promoters of pluripotency genes, morphology and proliferation properties which are similar to ESCs [201, 203, 204]. In addition, iPSCs and pluripotent cell hybrids are able to differentiate into the three embryonic germ layers *in vitro* and in teratomas [201, 203]. However, the levels of expression of mtDNA transcription and replication factors, number of mitochondria and mtDNA copies/cell or state of mitochondrial metabolic activity have not been investigated in either of these pluripotent type cells. Therefore, it remains to be determined whether these pluripotent cells also express the mitochondrial and mtDNA properties characteristic of pre-implantation embryos and undifferentiated ESCs.

As for SCNT, reprogramming a somatic cell through transcription-induced pluripotency and cell fusion is probably not limited to epigenetic changes at the nuclear level but also reprogramming of the cytoplasm. As demonstrated earlier, pluripotency seems to be associated with anaerobic metabolism, low levels of mtDNA transcription and replication and poorly developed mitochondria in ESCs [5–7, 152] and in pre-implantation embryos [126, 144, 145]. Therefore, reprogramming the somatic cell to acquire pluripotency to differentiate into the three embryonic germ layers is likely to require acquisition of less well-developed and more quiescent mitochondria. It is likely that reprogramming of nDNA results in patterns of expression of nDNA-encoded ETC subunits and mtDNA transcription and replication factors typical of undifferentiated ESCs. In addition, the continuous proliferation of the pluripotent cells in culture might lead to a decreased number of mitochondria and mtDNA copies/cell. This would result in reduced oxidative capacity and most likely drive the cells to activate anaerobic metabolism as the major source of ATP

production. Failure to reprogram mitochondria in pluripotent cells derived from SCNT, transcription-induced pluripotency or cell fusion might result in fully differentiated cells having inappropriate numbers of mitochondria and mtDNA which may lead to compromised cellular function.

Conclusions

A considerable amount of data has shown that human and mouse ESCs and the pluripotent cells of the mammalian embryos contain small and immature mitochondria, express low levels of ETC subunits, have low levels of oxygen consumption and limited capacity for aerobic production of ATP. Moreover, these mitochondrial properties seem to be conserved not only in pluripotent embryonic cells but also in adult stem cells suggesting that they might be important for the maintenance of both pluripotency and multipotency. It is likely that the derivation of ESCs through SCNT, induced pluripotency or pluripotent cell hybrids would require reprogramming of the somatic cell mitochondria to acquire properties similar to those mitochondria in early embryos and undifferentiated ESCs. As previously mentioned, failure to reprogram mitochondria would result in pluripotent cells containing inappropriate numbers of mitochondria and mtDNA copies/cell which is likely to result in failure to regulate mitochondrial biogenesis according to the metabolic requirement of the specific cell-type. Indeed, successful differentiation of embryonic cells *in vivo* or ESCs *in vitro* involves initiation of mtDNA transcription and replication, an increase in the number of mitochondria and upregulation of enzymes required for aerobic metabolism in order to fulfil the elevated ATP requirements of fully differentiated cells.

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